1.0 VECTOR OPERATIONS

LEARNING GOALS

Students will:

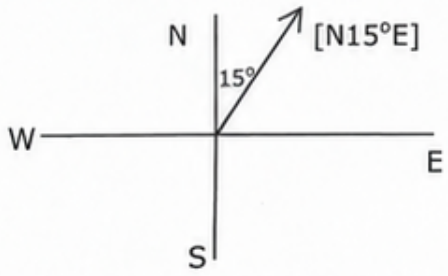
- Describe vectors graphically and mathematically
- Add and subtract vectors using mathematical methods

VECTORS

1. Write a definition for a vector: A quantity with magnitude and direction

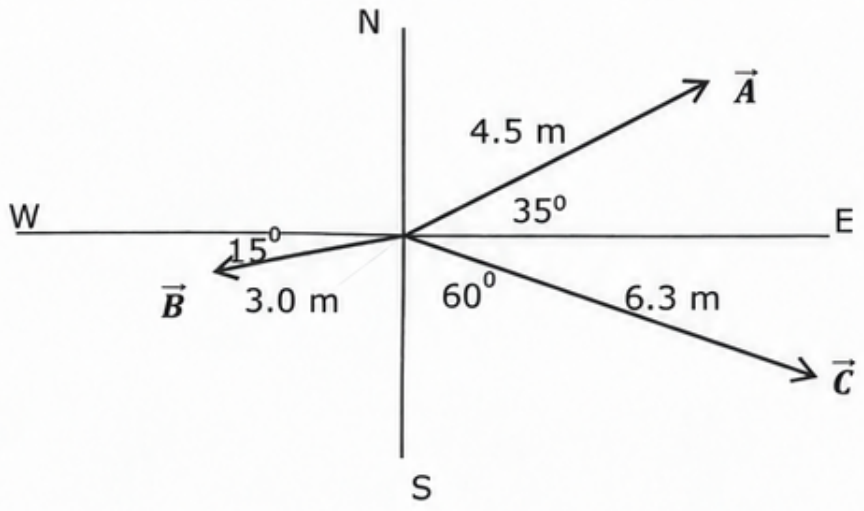
STEPS TO DRAWING VECTORS

1. Draw an x,y axis.
2. Measure the angle starting from the x-axis or based on the letter direction provided. A direction like [N15°E] is read "north, fifteen degrees east". This direction is found by starting at the north line and measuring 15° towards the east. Note: [N15°E] can also be written as [15° E of N]



WRITING VECTORS

Here are some vectors drawn in a horizontal plane (not to scale):



True bearing: use angle starting from north and going clockwise

magnitude [bearing]

ex. 6.8 m [050°]

a) State each vector in the form *magnitude [direction]* (e.g. 5.0 m [E25°N]):

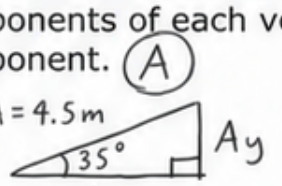
$\vec{A}$ 4.5 m [E 35°N]

$\vec{B}$ 3.0m [W15°S]

$\vec{C}$ 6.3 m [S60°E]

b) Each vector has components in two of the main directions. For example, **A** has a north component and an east component as it is in between north and east. Sketch the components of each vector and use trigonometry to find the magnitude of each component.

A



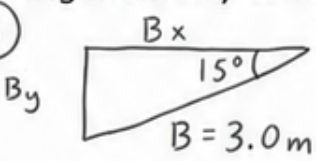
$\sin 35° = \frac{\vec{A}y}{\vec{A}}$

$Ay = 2.6\,m$

$\cos 35° = \frac{\vec{A}x}{\vec{A}}$

$Ax = 3.7\,m$

B



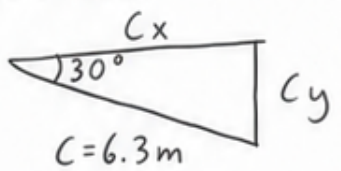
$\sin 15° = \frac{By}{B}$

$By = 0.78\,m$

$\cos 15° = \frac{Bx}{B}$

$Bx = 2.9\,m$

C



$\sin 30° = \frac{Cy}{C}$

$Cy = 3.2\,m$

$\cos 30° = \frac{Cx}{C}$

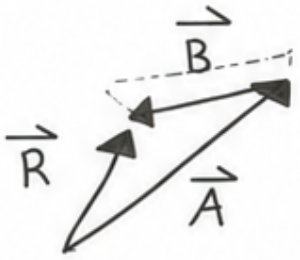
$Cx = 5.5\,m$

ADDING VECTORS

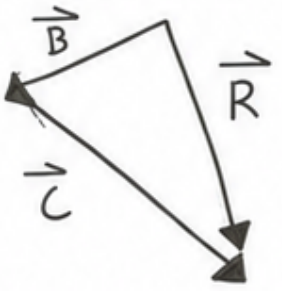
Recall from the video that when you add vectors, you draw them **tip to tail**. Then you draw the resultant from the beginning of the first vector to the tip of the last vector. To calculate the resultant, you must find its **magnitude** and **direction**. This will usually involve Pythagorean Theorem, trigonometry and/or geometry.

1. Draw a vector diagram showing the addition of vectors. No calculations required.

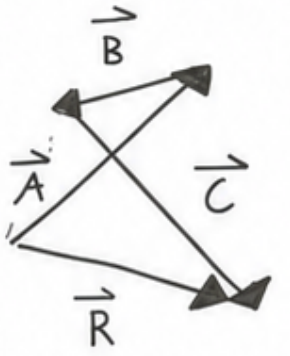
$\vec{A} + \vec{B}$



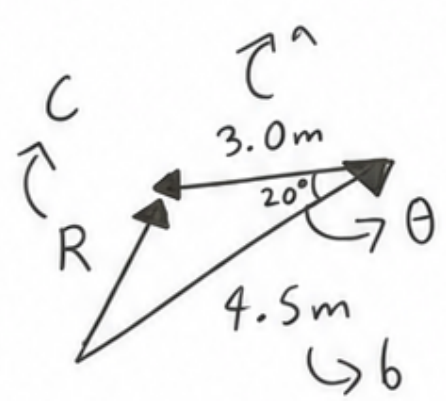
$\vec{B} + \vec{C}$



$\vec{A} + \vec{B} + \vec{C}$



2. The first two examples above are triangles. Use the sine and cosine laws to solve for the unknown side of $\vec{A} + \vec{B}$ (the side representing the sum). You will have to figure out the angle between the two known vectors first.



Original seperation angle: 160°
Internal angle and original seperatian angle are supplomentary
∴ 160° + θ = 180
θ = 20°

cosine Law

$C = \sqrt{a^2 + b^2 - 2ab\cos\theta}$

$C = 2.0\ m$

3. Repeat the additions you performed above by adding the components of the vectors. Also add all three vectors using the component technique. Draw a diagram showing components for each one.

$\vec{A}+\vec{B}$

$\vec{A}x = 3.7\,m$ →

$\vec{B}x = -2.9\,m$ ←

$\vec{R}x = \vec{A}x + \vec{B}x$

$\vec{R}x = 0.80\,m$

use $a^2+b^2=c^2$

$R^2 = Rx^2 + Ry^2$

$R = 2.0\,m$

$\tan\theta = \frac{Ry}{Rx}$

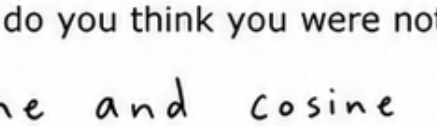
$\theta = 66^\circ$

$\vec{B}y = -0.78\,m$ ↓

$\vec{A}y = 2.6\,m$ ↑

$\vec{R}y = \vec{A}y + \vec{B}y$

$\vec{R}y = 1.82\,m$ ↑



$\vec{A}+\vec{B} = 2.0\,m\,[E66^\circ N]$

$\vec{A}+\vec{B}+\vec{C}$

$\vec{A}x = 3.7\,m$ →

$\vec{B}x = -2.9\,m$ ←

$\vec{C}x = 5.5\,m$ →

$\vec{R}x = \vec{A}x + \vec{B}x + \vec{C}x$

$\vec{R}x = 6.3\,m$

use. $a^2+b^2=c^2$

$R = \sqrt{Rx^2 + Ry^2}$

$R = 6.4\,m$

$\tan\theta = \frac{|Ry|}{|Rx|}$

$\theta = 12^\circ$

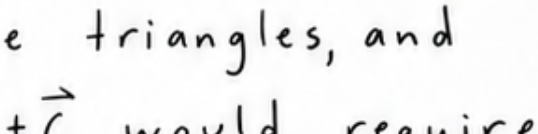
$\vec{B}y = -0.78\,m$ ↓

$\vec{A}y = 2.6\,m$ ↑

$\vec{C}y = -3.2\,m$ ↓

$\vec{R}y = \vec{A}y + \vec{B}y + \vec{C}y$

$\vec{R}y = -1.38\,m$



$\vec{A}+\vec{B}+\vec{C} = 6.4\,m\,[E12^\circ S]$

4. Why do you think you were not asked to do $\vec{A}+\vec{B}+\vec{C}$ using sine and cosine law?

Sine and cosine laws require triangles, and the path formed by $\vec{A}+\vec{B}+\vec{C}$ would require solving 2 triangles, which is more work.

SUBTRACTING VECTORS

The simplest way to subtract vectors is to add the negative.

EXAMPLE:

$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

$\vec{A} - \vec{B}$ = 4.5 m [E35°N] + (-3.0 m [W15°S])

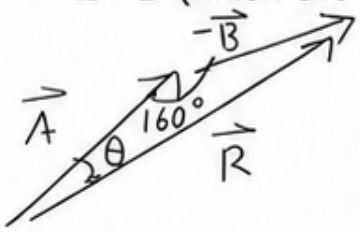
= 4.5 m [E35°N] + 3.0 m [E15°N]

Note that you are now adding $\vec{A}$ to the negative or opposite direction of $\vec{B}$.

Now you can perform the addition as you did above.

1. Evaluate the following. Draw a **diagram** for each one.

$\vec{A} - \vec{B}$ (finish the example)



Angle between $\vec{A}$ and $-\vec{B}$ = 20°

$-\vec{B} = 3.0\ [E15°N]$

$\vec{A} = 4.5m\ [E35°N]$

use cosine law

$c^2 = a^2 + b^2 - 2ab\cos C$

$R^2 = 4.5^2 + 3^2 - 2(4.5)(3)\cos(160°)$

$\vec{A} - \vec{C}$

$R^2 = 54.622\ m^2$

$R = \sqrt{54.622}$

$R = 7.39\ m$

$R = 7.4m$

Sine law to find θ

$\sin\theta / 3.0 = \sin 160° / 7.39$

$\theta = 7.98°$

$35° - 7.98° = 27°$ north of east

$\vec{A} - \vec{B} = 7.4m\ [E27°N]$

PRACTICE

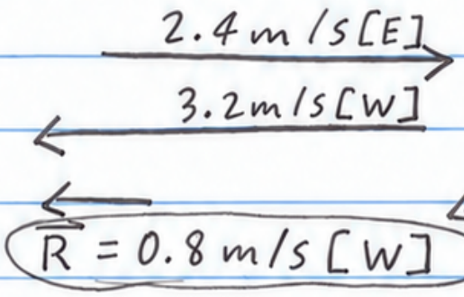
1. Do the following vector operations on the given velocities. Use whichever method you prefer, but make sure you practice **both** methods.
   a) 2.4 m/s [E] + 3.2 m/s [W]
   b) 2.4 m/s [E] + 3.2 m/s [N]
   c) 2.4 m/s [E] + 3.2 m/s [E45°S]
   d) 2.4 m/s [E] + 3.2 m/s [W35°N]
   e) 2.4 m/s [E35°N] + 3.2 m/s [W35°N]
   f) 2.4 m/s [E] - 3.2 m/s [W35°N]
   g) 2.4 m/s [E] + 3.2 m/s [W35°N] + 4.2 m/s [S10°W]
2. Optional Online Activity: ExploreLearning.com has a vector Gizmo you can play with.

Answers:

a) 0.8 m/s [W] b) 4.0 m/s [E53°N] c) 5.2 m/s [E26°S]
d) 1.8 m/s [W83°N] e) 3.3 m/s [W78°N] f) 5.3 m/s [E20°S] g) 2.5 m/s [S22°W]

Vectors (practice)

a) 2.4 m/s [E] + 3.2 m/s [W]



$\vec{R} = \vec{A} + \vec{B}$

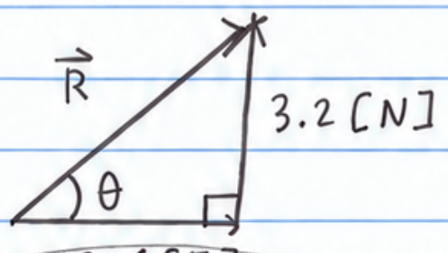
$Rx = 2.4\,m/s - 3.2\,m/s$

$Rx = -0.8\,m/s$

(negative is west here)

$\vec{R} = 0.8\,m/s\,[W]$

b) 2.4 m/s [E] + 3.2 m/s [N]



$9^2 + 6^2 = c^2$

$R = \sqrt{2.4^2 + 3.2^2}$

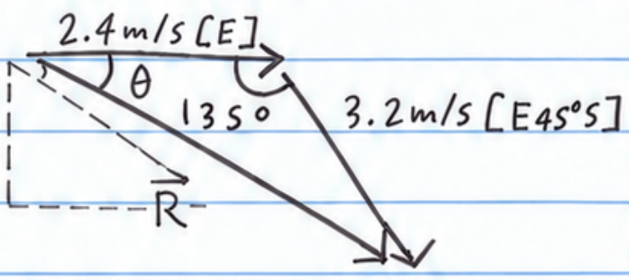
$R = 4.0\,m/s$

$\theta = \tan^{-1}(3.2/2.4)$

$\theta = 53°$

$\vec{R} = 4.0\,m/s\,[E\,53°N]$

c) 2.4 m/s [E] + 3.2 m/s [E 45° S]



use cos law

$c = \sqrt{a^2 + b^2 - 2ab\cos C}$

$R = \sqrt{2.4^2 + 3.2^2 - 2(2.4)(3.2)\cos 135°}$

$R = 5.183\,m/s$

$R = 5.2\,m/s$

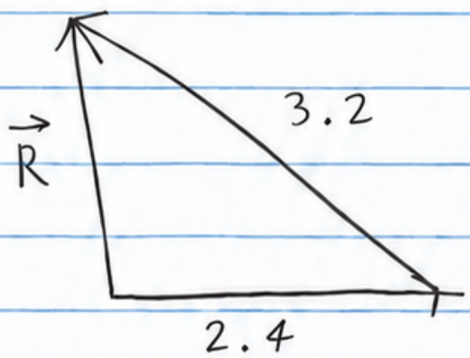
$\sin\theta / 3.2 = \sin 135° / R$

$\theta = 26°$ south of east

$\vec{R} = 5.2\,m/s\,[E\,26°S]$

Vectors Practice

d) 2.4 m/s [E] + 3.2 m/s [W 35° N]



$\vec{R}x = 2.4 - 3.2\cos 35°$

$\vec{R}x = -0.22$ m/s

$\vec{R}y = 3.2\sin 35°$

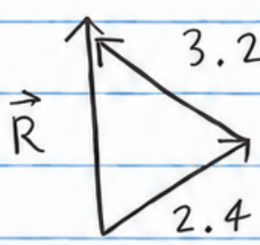
$\vec{R}y = 1.84$ m/s

$\theta = \tan^{-1}(|Ry|/|Rx|)$

$\theta = 83°$ north of west

$\vec{R} = 1.8$ m/s [W 83° N]

e) 2.4 m/s [E 35° N] + 3.2 m/s [W 35° N]



$\vec{R}x = (2.4 - 3.2)\cos 35°$

$\vec{R}x = -0.6553$ m/s

$\vec{R}x = -0.66$ m/s

$\vec{R}y = (2.4 + 3.2)\sin 35°$

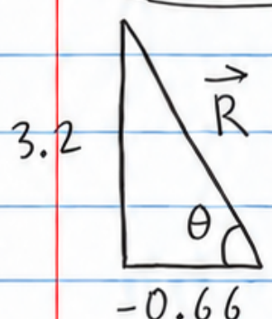
$\vec{R}y = 3.212$ m/s

$\vec{R}y = 3.2$ m/s

$R = \sqrt{Rx^2 + Ry^2}$

$R = 3.28$ m/s

$R = 3.3$ m/s

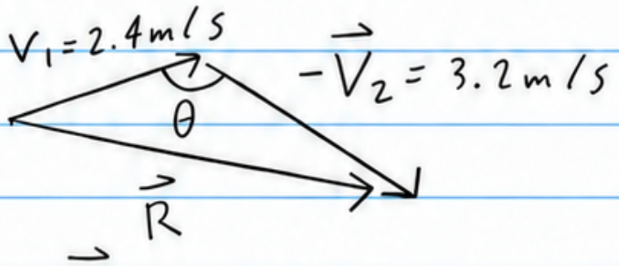


$\theta = \tan^{-1}(|Ry|/|Rx|)$

$\theta = 78°$ north of west

$\vec{R} = 3.3$ m/s [W 78° N]

f) 2.4 m/s [E] − 3.2 m/s [W35°N]

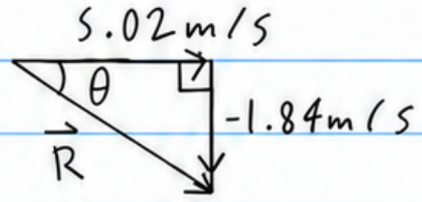


$\vec{R}x = 2.4 + 3.2 \cos 35°$

$\vec{R}x = 5.02 m/s$

$\vec{R}y = -3.2 \sin 35°$

$\vec{R}y = -1.84 m/s$



$R = \sqrt{Ry^2 + Rx^2}$

$R = 5.347 m/s$

$R = 5.3 m/s$

$\vec{R} = 5.3 m/s [E20°S]$

$\theta = \tan^{-1}\left(\frac{|Ry|}{|Rx|}\right)$

$\theta = 20°$ south of east

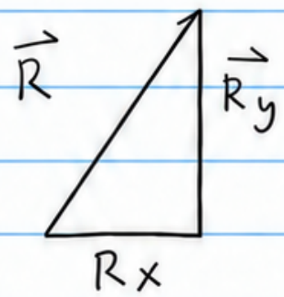
g) 2.4 m/s [E] + 3.2 m/s [W35°N] + 4.2 m/s [S10°W]

$Rx = 2.4 - 3.2 \cos 35° - 4.2 \sin 10°$

$= -0.95 m/s$

$Ry = 3.2 \sin 35° - 4.2 \cos 10°$

$= -2.3 m/s$



$R = \sqrt{Rx^2 + Ry^2}$ $R = 2.5 m/s$

$\theta = \tan^{-1}(Ry / |Ry|)$

$= 22°$ west of south

$\vec{R} = 2.5 m/s [S22°W]$

